# ■ EcoRoute — Comprehensive Technical & Staff Presentation Documentation

Official AI-Powered E-Waste Management & Recycling Marketplace | CPCB & MoEFCC Compliant

## 1. Executive Overview

**EcoRoute** is an enterprise-grade, AI-powered e-waste management platform designed to address India's growing electronic waste challenge. It provides a seamless bridge between **Citizens (Sellers)** and **Certified Recyclers (Buyers)** in compliance with Ministry of Environment, Forest and Climate Change (MoEFCC) and Central Pollution Control Board (CPCB) standards.

**Key Objectives:**

- **AI Computer Vision Scrap Valuation:** Enables citizens to scan e-waste items for instant CPCB-compliant price valuations and Green Coin rewards.
- **Doorstep Pickup & CPCB Token Generation:** Streamlines recycling logistics with scannable QR tokens, scheduled pickup windows, and Google Maps routing.
- **Green Coin Rewards & Utility Bill Rebates:** Citizens earn Green Coins (1 Coin = ■1) redeemable for instant UPI cash transfers, Electricity (EB) bill discounts, or Water bill waivers.
- **Strict Role-Based Security:** Isolated login portals and tab validation for Citizens and Recyclers to prevent fraud.

## 2. Complete Technology Stack

| Layer | Technology Used | Description / Function |
| --- | --- | --- |
| Framework | Next.js 16 (App Router) | Full-stack React 19 framework providing Server Components and API routes. |
| Language | TypeScript 5 & Node.js 18+ | Type-safe execution across client components and backend API services. |
| Styling Engine | Tailwind CSS v4 | Utility-first CSS engine with custom government color tokens (#003366, #2E7D32). |
| Animations | Framer Motion 12 | GPU-accelerated micro-animations and page entry transitions. |
| Database & Auth | Supabase (PostgreSQL) + Prisma | Enterprise database storing users, e-waste listings, and pickup schedules. |
| Backend REST API | Express.js & Next.js API Routes | Dual REST API layer handling JWT auth, valuation, and pickup tokens. |

## 3. Full-Screen Background Video Engine Breakdown

EcoRoute features an immersive full-screen background video engine on the homepage and portal dashboards. Here is how it functions technically:

### A. 100% Reliable Autoplay Compliance

Modern browsers (Chrome, Safari, Edge) strictly block videos from playing automatically if audio is enabled. EcoRoute sets explicit attributes autoPlay, muted, loop, playsInline, preload='auto', and uses a React useRef hook to force v.muted = true and v.play() on DOM mount.

### B. Dynamic Aspect-Ratio Scaling (object-fit: cover)

The video element uses CSS position: fixed / absolute covering 100vw × 100vh with object-fit: cover. This ensures the video fills desktop monitors, laptops, tablets, and smartphones without stretching.

### C. Hardware-Accelerated Cinematic Zoom (slowZoom)

To prevent a static feel, a 20-second CSS keyframe animation applies a continuous scale(1.0) to scale(1.08) zoom effect. Using will-change: transform ensures GPU hardware acceleration for smooth 60 FPS performance.

```
@keyframes slowZoom {
from { transform: scale(1.0); }
to { transform: scale(1.08); }
}
.hero-video-zoom {
animation: slowZoom 20s linear infinite alternate;
will-change: transform;
```

```
}
```

### D. Dark Overlay & Glassmorphic Card Readability

A dark gradient overlay (opacity 50–75%) sits directly over the video (z-index: 1). Content panels use Glassmorphism (backdrop-filter: blur(20px)) so that all text passes **WCAG AAA accessibility contrast standards**.

### E. Battery & GPU Resource Saver

A visibilitychange event listener automatically pauses the background video when the user switches browser tabs or minimizes the window, saving laptop battery and GPU cycles.

# 4. Staff Presentation Script (Step-by-Step Outline)

Use this structured script when presenting the website to management, staff members, or stakeholders:

### Step 1: Introduction & Purpose (1 Minute)

*"Good morning team. Today I am presenting **EcoRoute**, our official AI-powered E-Waste Management & Recycling Portal built according to Central Pollution Control Board (CPCB) standards. EcoRoute makes e-waste disposal effortless for citizens while creating a verified digital marketplace for certified recyclers."*

### Step 2: Showcasing the Landing Page & Video Engine (2 Minutes)

*"As you see on the homepage, we have implemented an immersive full-screen background video engine. Notice how smoothly the background video plays with a subtle cinematic zoom. We engineered this with automatic browser policy compliance, GPU hardware acceleration, and glassmorphic overlays to ensure 100% legibility and high accessibility contrast."*

### Step 3: Citizen Flow & AI Valuation (3 Minutes)

*"When a citizen logs in under the Citizen Portal:*
*1. They navigate to **Upload E-Waste**.*
*2. Our AI Computer Vision auto-identifies the electronic device and provides an immediate CPCB valuation price.*
*3. Upon submitting, the listing is posted to the marketplace, and the user earns **Green Coins**.*
*4. In the **Rewards** tab, citizens can redeem Green Coins 1:1 for instant UPI cash, Electricity bill discounts, or Water bill waivers."*

### Step 4: Recycler Marketplace & CPCB Token (3 Minutes)

*"Now switching over to the Recycler / Buyer Portal:*
*1. Certified recyclers browse available listings across their district or state.*
*2. They click **Intimate Pickup & Cash** to schedule a collection date and payment method.*
*3. The platform automatically generates a CPCB-compliant digital purchase token complete with live Google Maps directions to the seller's doorstep."*

### Step 5: Security & Conclusion (1 Minute)

*"Finally, the system enforces strict role isolation—blocking cross-role logins—and features a 1-hour session timeout for government compliance. The platform is fully responsive and supports 6 official Indian languages."*